

Vineet Agarwal

240-353-9811 — vineetagarwal540@gmail.com — vineetagarwal54 — Github — Portfolio

Summary — Software engineering graduate student with 2+ years of production experience building multi-agent AI systems and applied LLM products. Ships RAG pipelines, agent orchestration and evaluation, and GenAI automation that run in production for real users, backed by hands-on LLM inference optimization and a full-stack foundation that treats AI as one reliable component inside a larger system.

Skills

Applied AI multi-agent systems, RAG with cited retrieval, agent evaluation and scoring, LangChain, OpenAI APIs, Claude, prompt engineering, embeddings, vector search

LLM Inference & Optimization speculative decoding, vLLM, SGLang, TensorRT-LLM, quantization (INT4, FP8), KV cache, PyTorch, HuggingFace, on-device (llama.rn)

GenAI Automation Microsoft Copilot, Power Automate, Azure Functions (Python), SharePoint, Microsoft Graph API

Languages Python, JavaScript, TypeScript, C++, SQL, Go

Frontend React, Next.js, React Native, Redux Toolkit

Backend & Cloud FastAPI, Node.js, Express, REST APIs, AWS (Lambda, API Gateway, DynamoDB, SQS), Docker, Kubernetes, CI/CD

Databases PostgreSQL, MongoDB, Redis, FAISS, Pinecone

Core Engineering Data Structures & Algorithms (C++), OOP, System Design, Microservices, Agile/Scrum

Experience

ServBeyond Solutions

Jun 2026 – Present

Enterprise AI & Platform Intern

- Shipped a multi-agent RAG proposal assistant used daily by 20+ business development staff, auto-vectorizing every document on insert into a version-controlled SharePoint archive, with a Claude drafting agent and a separate evaluation agent that scores each RFI draft against the solicitation before reviewer handoff.
- Engineered a company-wide AI meeting autopilot for 100 employees, chaining 5 Power Automate flows and Microsoft Graph API to convert Teams Copilot recaps into routed Jira tickets, mapping context to catch cross-meeting follow-ups and update existing tickets instead of duplicating, cutting up to 2 hours of manual entry per meeting.
- Built a natural-language USAspending research agent whose query planner parses each prompt, routes to the correct API, and retrieves 1000+ contracts across any year range, generating Excel dashboards saved to SharePoint and delivered through Outlook and Teams.

Runara.ai

March 2026 – May 2026

Software Engineer Intern

- Implemented speculative decoding from scratch in PyTorch and HuggingFace across two GPUs, applying the rejection-sampling acceptance criterion and establishing the draft-to-target cost ratio under which the technique yields 2 to 3x inference throughput on large target models.
- Benchmarked Qwen3-Coder-480B across vLLM, SGLang, and TensorRT-LLM on Blackwell GPUs, profiling TTFT, throughput, VRAM, and KV cache growth, and caught an FP8 versus 4-bit checkpoint mismatch that prevented wasted GPU hours.

Xelpmoc Design and Tech Limited

Nov 2022 – Apr 2024

Software Engineer

- Reduced API response time from 75s to under 10s and raised throughput 4x by refactoring SQL joins, indexing high-traffic tables, and adding a Redis caching layer.
- Architected RESTful backend services for a tourism platform, cutting production defects 40% and accelerating feature delivery 25%.

IIIT Hyderabad

Jun 2022 – Aug 2022

Software Intern

- Automated multilingual data extraction across 5+ regional languages with Python and Selenium, improving NLP classification accuracy 15%.

Education

University of Maryland, College Park

Jan 2025 – Dec 2026 (Expected)

Master of Engineering, Computer Software Engineering — Minor in Cloud Engineering

GPA: 3.7

Projects

RepoResearchAI : Multi-Agent Code Analysis Platform (Github)

- Designed a multi-agent platform with AutoGen, LangChain, and FAISS vector search where specialized agents (SDE, PM, QA) produce architecture insights, API docs, and schema analysis using GPT-4 with RAG over codebase embeddings.

Locra: On-Device Multimodal AI, Android (Github)

- Developed an Android-first multimodal AI assistant in React Native, TypeScript, and Expo that runs a quantized Qwen3-VL-2B vision-language model fully on device via llama.rn, pairing on-device Whisper speech-to-text with a Kotlin native module that enforces model integrity and a hard offline guarantee.